

Learn, Denoise and Discover:

AI-powered Denoising for Electron Microscopy

Motivation: Studying catalysis

90% of all manufactured goods involve catalytic processes somewhere in their production chain

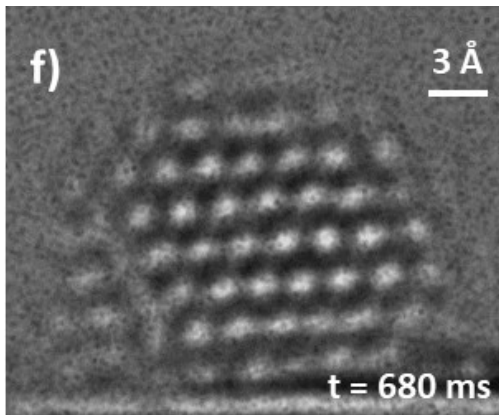
Important for energy, pharmaceuticals, materials, transport, and environmental applications

Nanoparticle-based catalysts are crucial

To understand functionality, we need to observe changes in the atomic structure of nanoparticles

Challenge: Atoms are tiny!

Order of magnitude: $1 \text{ \AA} = 10^{-10} \text{ m}$

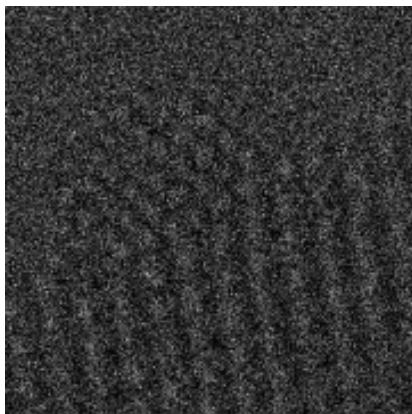


Electron microscopy enables visualization of the atomic structure by shooting electrons at the material

Challenge: Atomic structure changes fast!

We need to acquire data as **fast** as possible

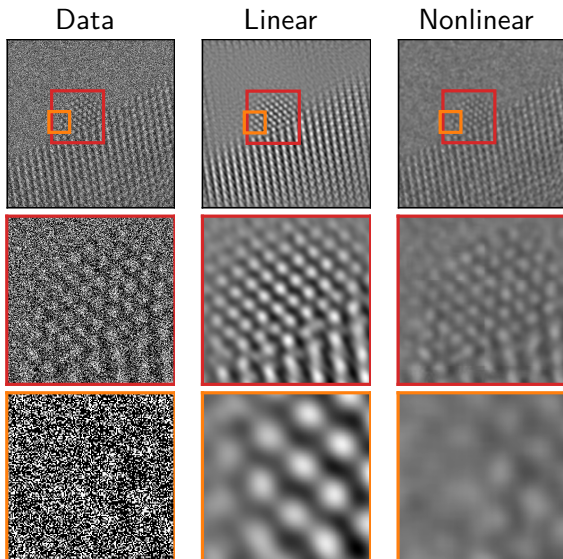
Acquisition at a timescale of **milliseconds** $\implies$ Weak signal



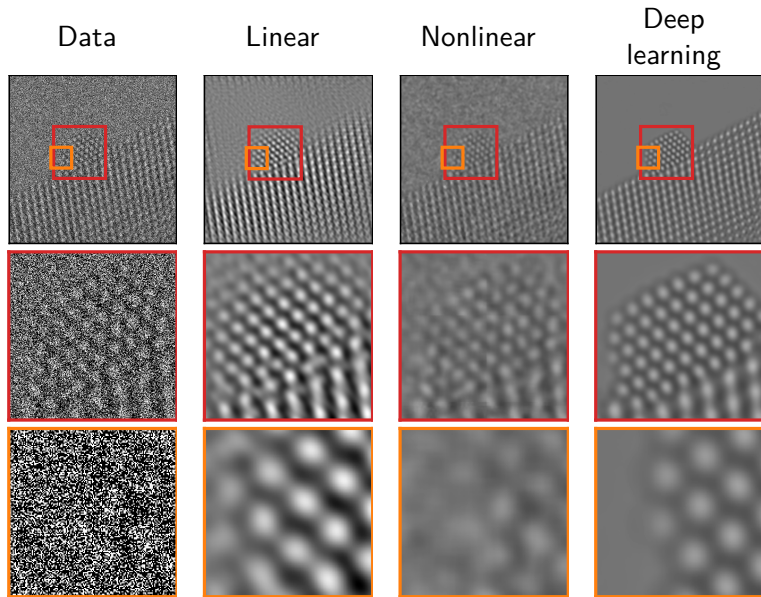
Possible solution: Higher electron dose $\implies$ **destroys material**

We need to **denoise**!

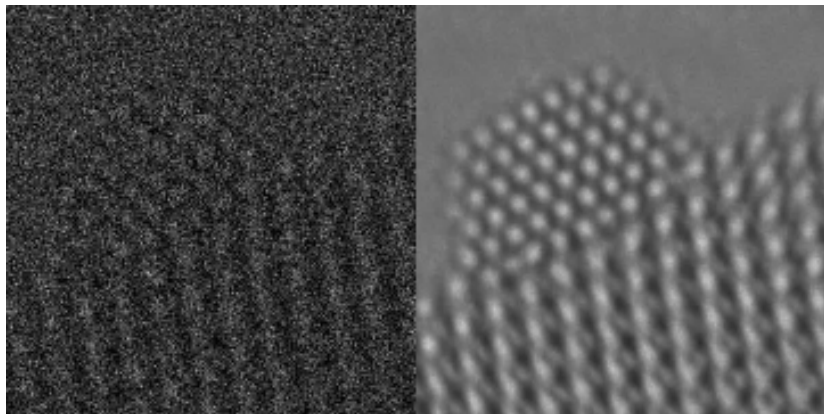
Traditional methods



AI-powered denoising



AI-powered denoising

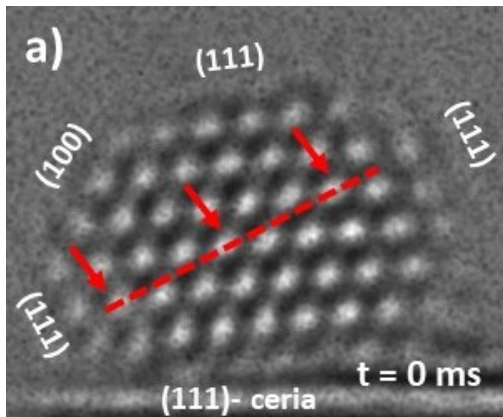


AI-powered denoising reveals unobserved dynamics

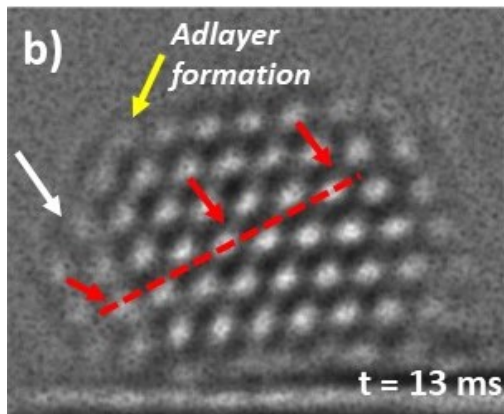
Visualizing nanoparticle surface dynamics and instabilities enabled by deep denoising. **Science** 387 (6737) pp:949-954

Crozier PA, Leibovich M, Haluai P, Tan M, Thomas AM, Vincent J, Mohan S, Marcos Morales A, Kulkarni SA, Matteson DS, Wang Y, Fernandez-Granda C

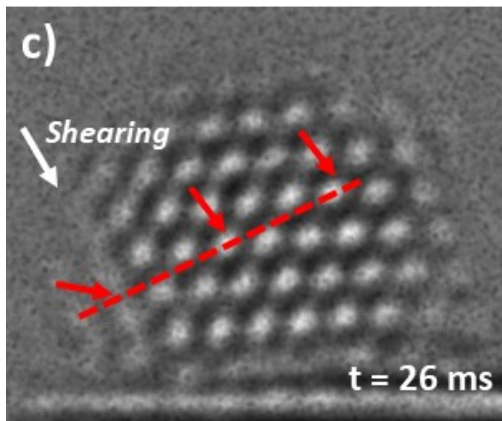
Shearing (displacement of atomic plane)



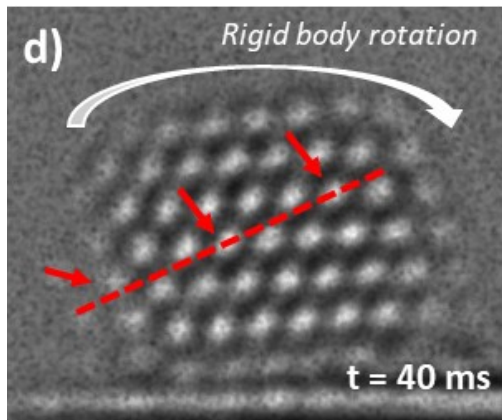
Shearing (displacement of atomic plane)



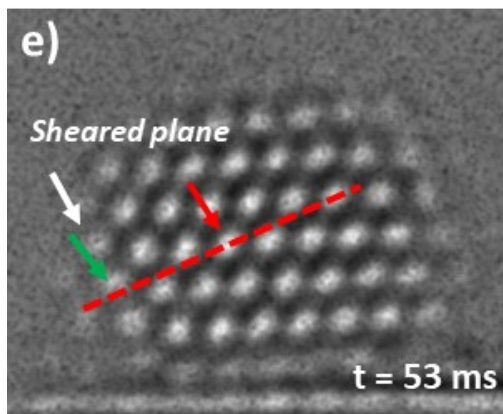
Shearing (displacement of atomic plane)



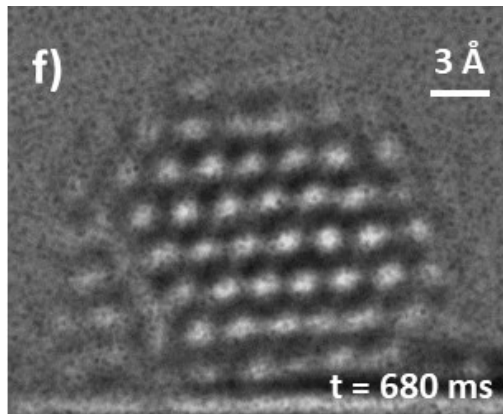
Rigid rotation



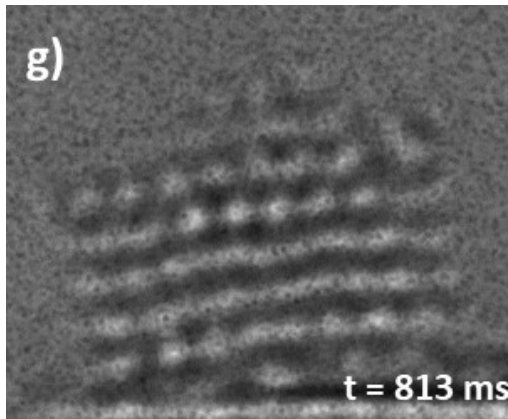
Rigid rotation



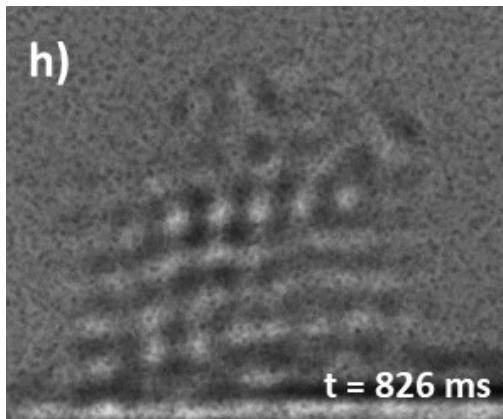
Destabilization (fluxionality)



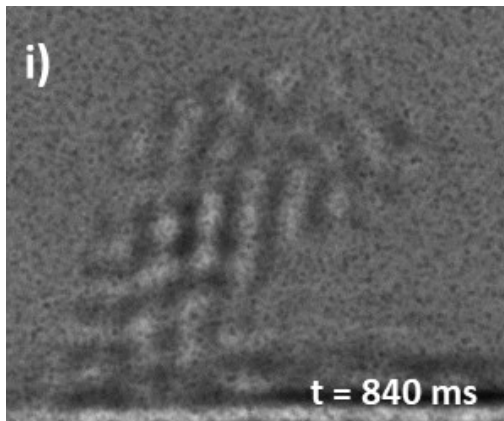
Destabilization (fluxionality)



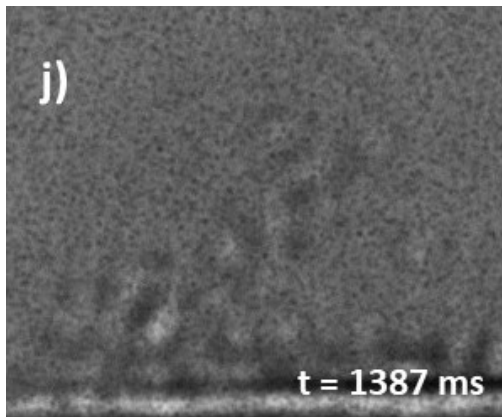
Destabilization (fluxionality)



Destabilization (fluxionality)



Destabilization (fluxionality)



Plan

Supervised learning

Unsupervised learning

Semi-supervised learning

Evaluation

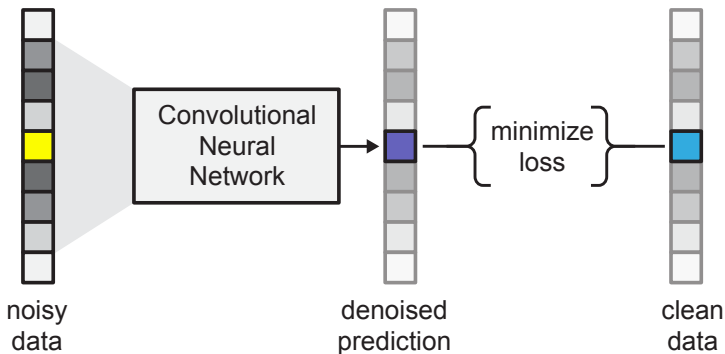
Supervised learning

Unsupervised learning

Semi-supervised learning

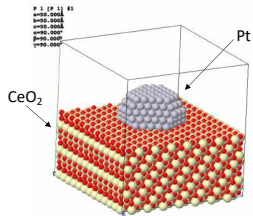
Evaluation

Supervised learning



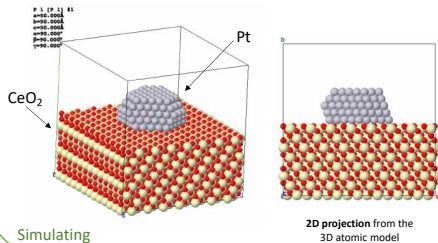
Problem: Requires access to clean - noisy pairs

Simulation-based denoising

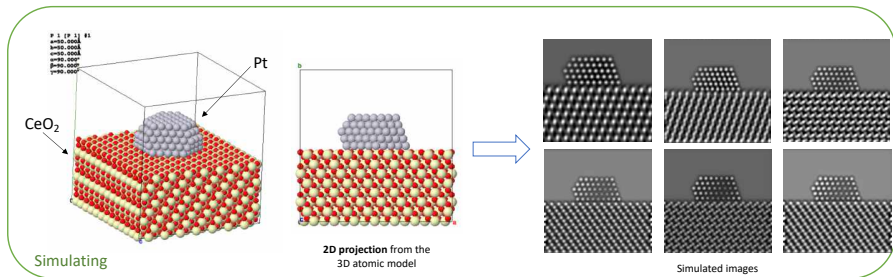


Simulating

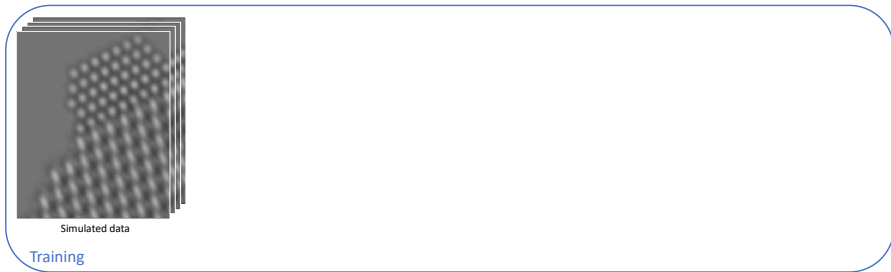
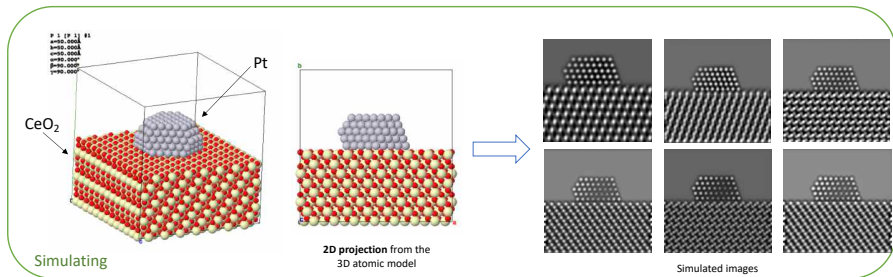
Simulation-based denoising



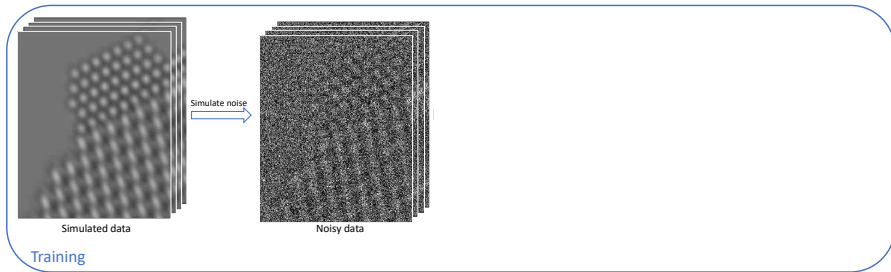
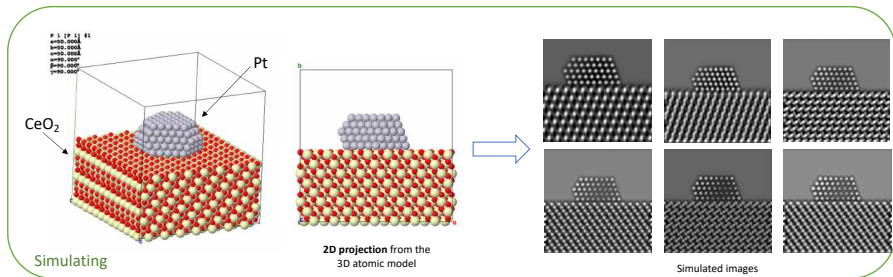
Simulation-based denoising



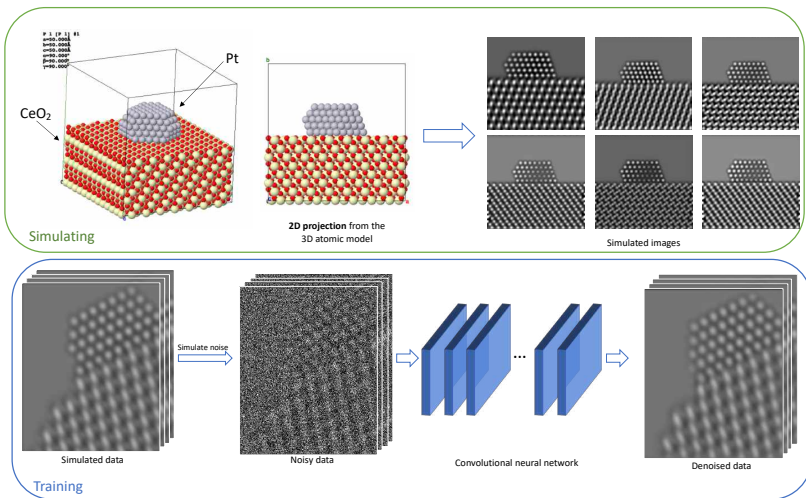
Simulation-based denoising



Simulation-based denoising

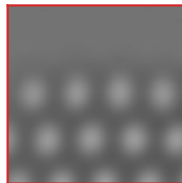
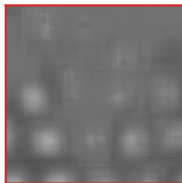
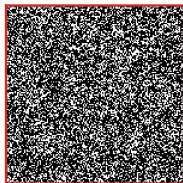
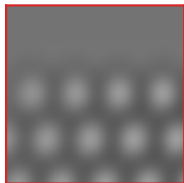
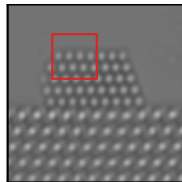
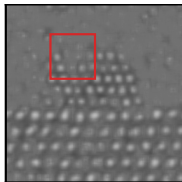
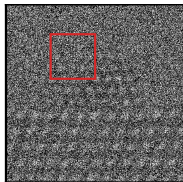
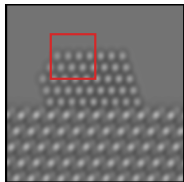


Simulation-based denoising



Deep denoising for scientific discovery: A case study in electron microscopy. Mohan et al. 2022

Lesson 1: Field of view



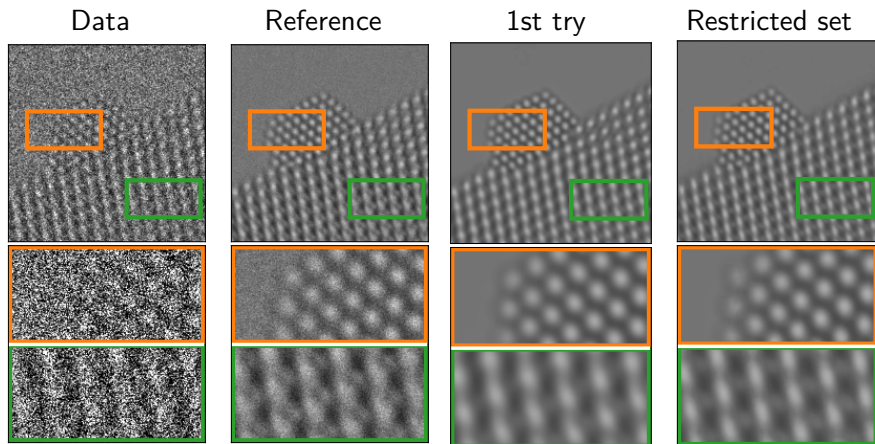
Simulated
clean image

Noisy image

DnCNN

UNet

Lesson 2: Design of training set



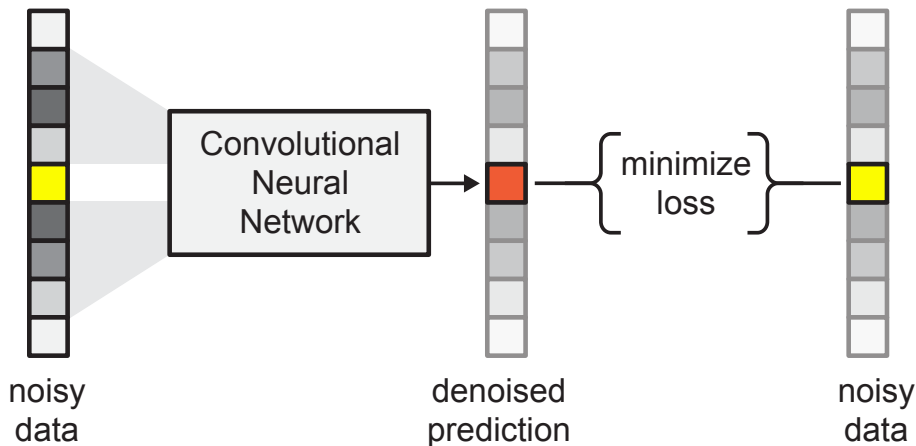
Supervised learning

Unsupervised learning

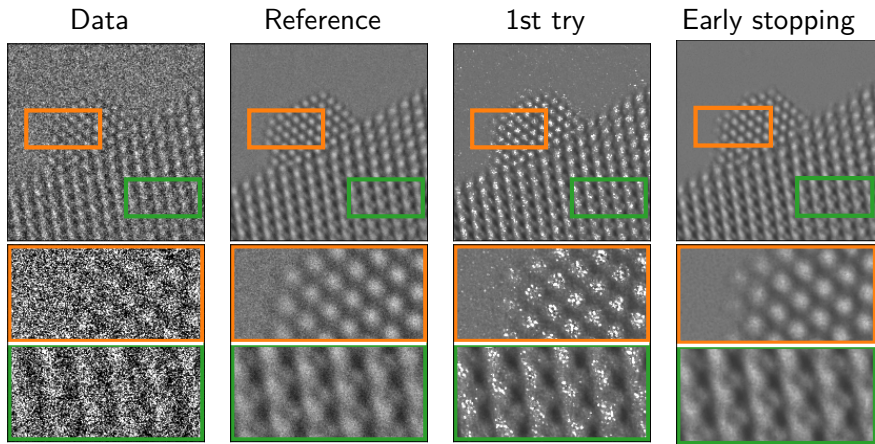
Semi-supervised learning

Evaluation

Blindspot denoising



Lesson: Unsupervised validation is crucial



Unsupervised Deep Video Denoising. Sheth et al. 2021

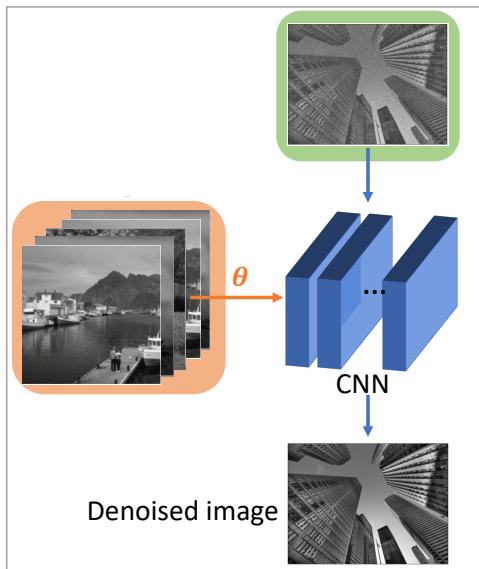
Supervised learning

Unsupervised learning

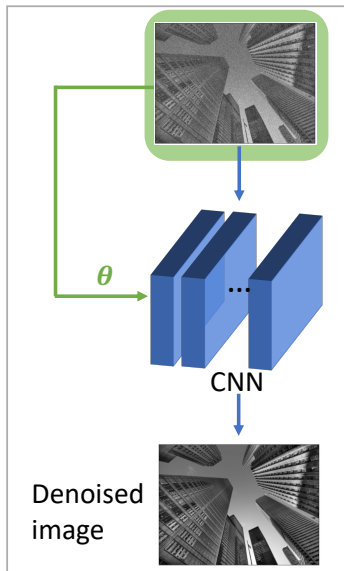
Semi-supervised learning

Evaluation

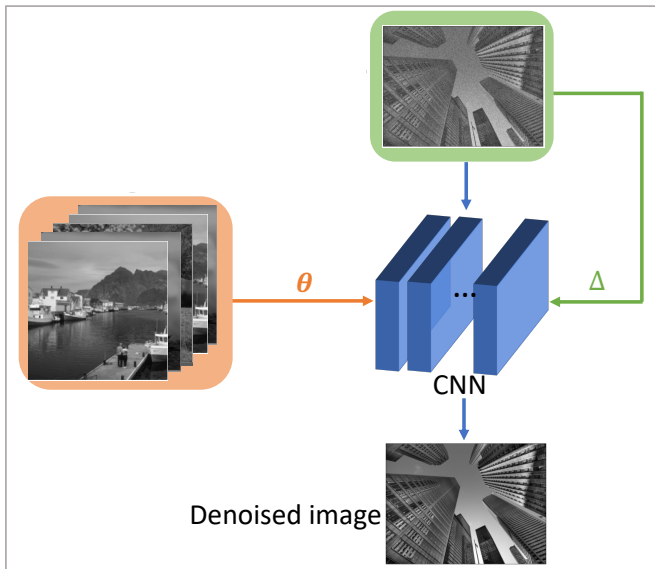
Supervised learning



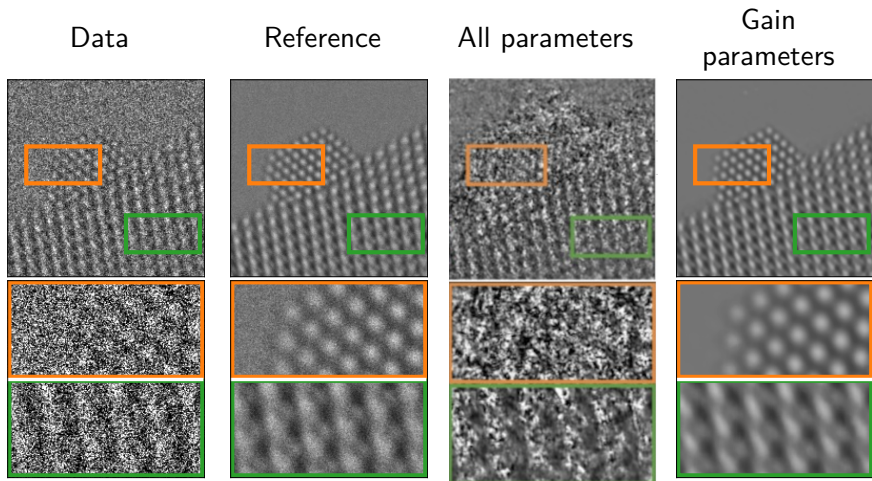
Unsupervised learning



Semi-supervised learning



Lesson: Cannot finetune all parameters



Adaptive denoising via gaintuning. Mohan et al. 2021

Supervised learning

Unsupervised learning

Semi-supervised learning

Evaluation

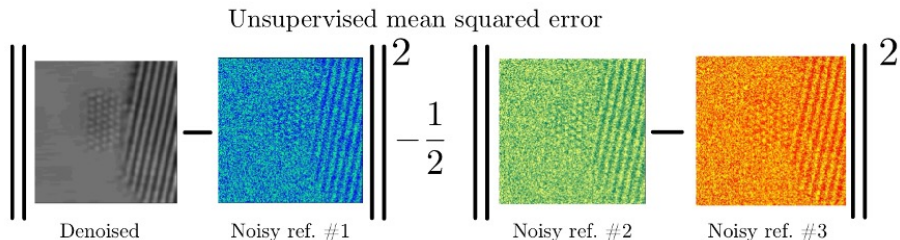
Supervised evaluation is not possible

Supervised mean squared error

$$\left\| \begin{array}{c} \text{Denoised} \end{array} - \begin{array}{c} \text{Clean} \end{array} \right\|_2^2$$

The diagram illustrates the supervised mean squared error (MSE) calculation. It shows two grayscale images side-by-side, separated by a minus sign. The left image is labeled 'Denoised' and the right image is labeled 'Clean'. Both images contain a grid of small squares and a set of diagonal lines. The 'Denoised' image is slightly blurred compared to the 'Clean' image. The entire expression is enclosed in a large norm symbol, $\left\| \cdot \right\|_2^2$, indicating the squared L2 norm of the difference between the two images.

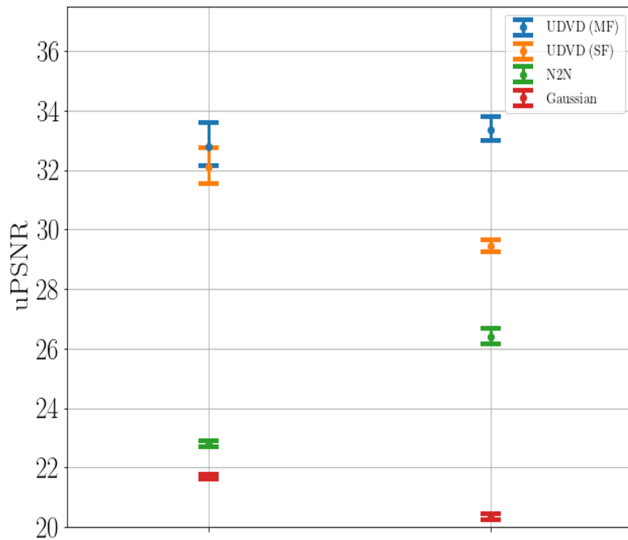
Unsupervised evaluation



Unbiased consistent estimator of MSE (assuming noise is independent across frames)

Evaluating unsupervised denoising requires unsupervised metrics.
Marcos-Morales et al. 2023

Unsupervised evaluation



Beyond MSE

MSE may not capture scientifically-meaningful structure

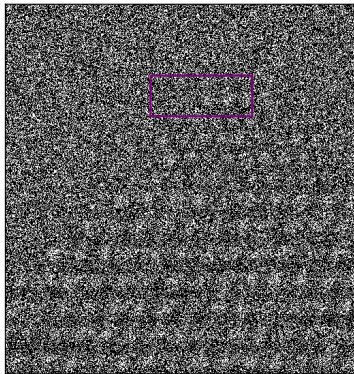
For example, presence or absence of atoms

Idea: Estimate *likelihood* of observed data given detected atoms

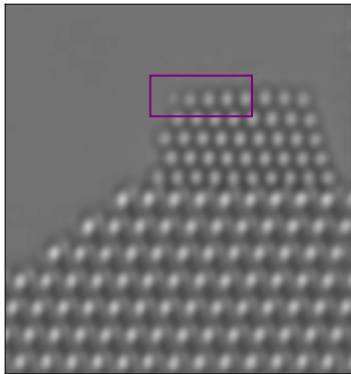
Deep denoising for scientific discovery: A case study in electron microscopy. Mohan et al. 2022

Beyond MSE

Data

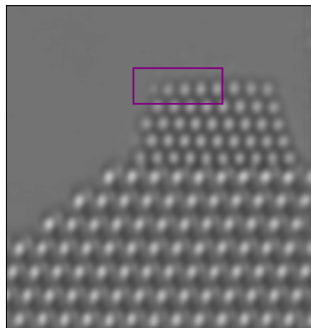


Denoised image

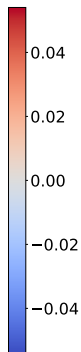
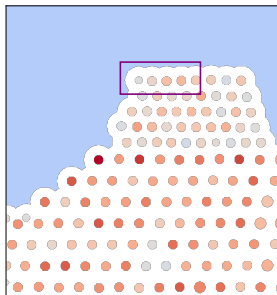


Likelihood map

Denoised image



Likelihood map



Conclusion

AI-powered denoising reveals hidden structure in electron microscopy data

Supervised, unsupervised and semi-supervised methods work, but overfitting is always looming!

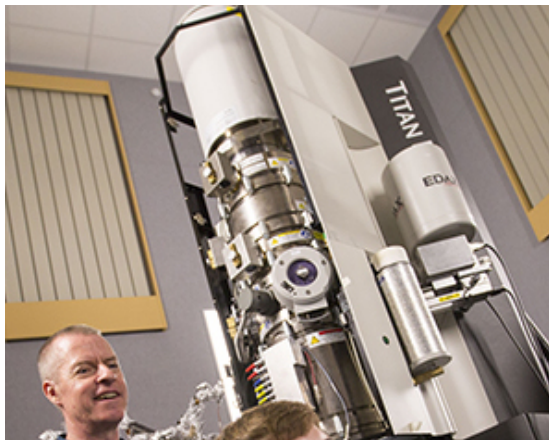
Evaluation is a major challenge

For more information

Learn, Denoise and Discover: A Guide to Deep Denoising with an Application to Electron Microscopy

Sreyas Mohan, Kangning Liu, Peter Crozier,
Carlos Fernandez-Granda

For more information



Crozier lab at Arizona State University

Acknowledgements

We are very grateful for the support of the [National Science Foundation](#) through grants OAC-2104105 and OAC-2103936